AIMS-Rwanda

Algebra and Cryptography- First assignment

Exercice 1. The following 5 questions are independent:

1. Prove that 13 divides $2^{70} + 3^{70}$.
2. Compute $\text{pgcd}(2^a - 1, 2^b - 1)$ for any a and b natural numbers.
3. Evaluate $\gcd(9n + 4, 2n - 1)$ where n is a natural number.
4. Let n be a natural number. Show that if $2^n + 1$ is a prime number, the integer n must be a power of 2. Prove by a counterexample that the converse is false.
5. Let p be an odd prime number. Consider two integer a and b such that p does not divide a nor b, but divides $a^2 + b^2$. Show that

$$p \equiv 1 \bmod 4.$$

Exercice 2. Bob intercepts from Alice the following encrypted message:

$$\begin{array}{l}[4278499682407590072284949786397750818809 \\ 4983082501366735895427485430308066299941 \\ 9252881053429437432710248374797072252555 \\ 9502432880041425490721735678390622574 0]\end{array}$$

Alice knows that Bob uses RSA cryptosystem and his public key is $(12398737, n)$ where

$$n = 956331992007843552652604425031376690367$$

Knowing that Alice and Bob agreed to use RSA cryptosystem to communicate in secret, each message consist of a single letter which is encoded as: Space $= 00, A = 11, B = 12, \cdots, Z = 36$, which message did Alice sent to Bob?

Exercice 3.
Create your own public key and private key for RSA cryptosystem. The two prime numbers must have 600 digits and they have to be safe prime numbers. Then, Set up your own RSA cryptosystem. Demonstrate how a message addressed to you can be encrypted and how you can decrypt it using your private key.